COHOMOLOGICAL DESCENT HOLDS FOR THE COMPARISON OF THE QC-FPQC AND TOPOLOGICAL REALISATIONS OF A LOCALLY QUASI-COMPACT SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction from a complex to a single sheaf by Leray's acyclicity lemma, the rewriting of the composite pushforward, and the degeneration of the relative Leray spectral sequence with the two vanishing inputs.

Editorial note. The label lemma-cohomological-descent-LC names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The comparison morphism a_X = epsilon_X o pi_X and its constituents. Same chapter, the morphisms of topoi epsilon_X and pi_X whose composite is a_X, and the description of the pullback along pi_X. Supplies the functors whose derived pushforwards are shown to vanish above degree zero.
- Vanishing of the higher pushforwards of the two constituents. Same chapter, the V-C statement giving the vanishing for epsilon_X and the collected Zariski properties giving R^j pi_{X,*} pi_X^{-1}F = 0 for j > 0. Supplies the two vanishing inputs feeding the relative Leray spectral sequence.
- Leray's acyclicity lemma and the relative Leray spectral sequence. Derived Categories, the acyclicity lemma, and Same chapter, the relative Leray spectral sequence for a composite. Supplies the reduction from a complex to a single sheaf and the vanishing of the higher direct images of a_X.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Prerequisite pointer invoked at proof line(s) 8109; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0DCY: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8049–8049. Prerequisite pointer invoked at proof line(s) 8119; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DCU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7770–7770. Prerequisite pointer invoked at proof line(s) 8122; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09X3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7675–7675. Prerequisite pointer invoked at proof line(s) 8114; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01F6: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2367–2367. Prerequisite pointer invoked at proof line(s) 8118; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be an object of LC_{qc}. For $K \in D^+(X)$ the map*

$$K \longrightarrow Ra_{X,*}a_X^{-1}K$$

is an isomorphism with $a_X : Sh(LC_{qc}/X) \to Sh(X)$ as above.

Proof. We first reduce the statement to the case where K is given by a single abelian sheaf. Namely, represent K by a bounded below complex $\mathcal{F}^\bullet$. By the case of a sheaf we see that $\mathcal{F}^n = a_{X,*}a_X^{-1}\mathcal{F}^n$ and that the sheaves $R^q a_{X,*}a_X^{-1}\mathcal{F}^n$ are zero for $q > 0$. By Leray's acyclicity lemma (Derived Categories, Lemma 015E) applied to $a_X^{-1}\mathcal{F}^\bullet$ and the functor $a_{X,*}$ we conclude. From now on assume $K = \mathcal{F}$.

By Lemma 09X3 we have $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$. Thus it suffices to show that $R^q a_{X,*}a_X^{-1}\mathcal{F} = 0$ for $q > 0$. For this we can use $a_X = \epsilon_X \circ \pi_X$ and the Leray spectral sequence Lemma 01F6. By Lemma 0DCY we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$ and $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$. By Lemma 0DCU we have $R^j\pi_{X,*}(\pi_X^{-1}\mathcal{F}) = 0$ for $j > 0$. This concludes the proof. □